

ROSENDO ADRIAN DE LOS RIOS MORENO

Robotics Engineer | Autonomous Systems · Computer Vision · Embedded Systems

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EDUCATION

B.Eng. in Robotics and Digital Systems Engineering

Tecnologico de Monterrey

Jun 2026

Monterrey, Mexico

GPA: 91/100 Relevant Coursework: Advanced Embedded Systems, ROS1/ROS2, Computer Vision, Artificial Intelligence, Autonomous Systems

EXPERIENCE

Research Intern — Robotic Teleoperation Systems

AIST Tokyo Waterfront, National Institute of Advanced Industrial Science and Technology

Aug 2025 – Feb 2026

Tokyo, Japan

- Reduced obstacle collision risk by implementing **ESDF-based assistive braking** on a **UR5e** arm, achieving **<6 cm** safety threshold at **40–70 ms braking latency**, by building a real-time **TSDF/ESDF** volumetric reconstruction pipeline in **ROS2** from Intel RealSense D435 depth data.
- Enabled live operator feedback at **15–18 Hz** with **140–210 ms end-to-end latency** by integrating a CPU **TSDF integrator** (Open3D), **DBSCAN** object clustering, and a **ROS-TCP-Connector** Unity bridge into a single teleoperation stack.

INDUSTRY COLLABORATION PROJECTS

Embedded Systems — Autonomous Tractor Navigation

John Deere · Socio-Formador Project, Tecnologico de Monterrey

Aug – Dec 2024

Monterrey, Mexico

- Delivered a functional navigation prototype by developing **C/C++ firmware** for **STM32** microcontrollers under **FreeRTOS**, integrating **IMU** sensors via **SPI/I2C** and motor control via **CAN/UART** across a full-semester industry collaboration.

Robotics Engineer — ROS2 Autonomous Mobile Robot

Manchester Robotics · Socio-Formador Project, Tecnologico de Monterrey

Feb – Jun 2024

Monterrey, Mexico

- Achieved real-time autonomous navigation by building a full **ROS2** control architecture in **C++/Python** that fused **YOLO**-trained vision with reactive planning, validated through simulation-based unit testing in **Gazebo**.

PROJECTS

Puzzlebot AMR — Autonomous Forklift Robot

Feb – Jun 2026

- Delivered a full autonomous mobile robot for pallet handling by developing an **HMM**-based voice command pipeline (11 commands, 93.6% accuracy), **ArUco** marker localization, **ICP** + log-odds **SLAM**, **EKF** localization in **C++/Eigen**, and **A*** + Bug1 navigation over **ROS2** with **LiDAR** sensing.
- Enabled remote monitoring and simulation by building a **Flask/SocketIO** web dashboard with live telemetry and a **Gazebo** digital twin; deployed mission control via a **YASMIN** state machine managing the full pick-and-place sequence.

Advanced Oil Pan Tracking & Manipulation

Feb – May 2025

- Achieved **2.8 cm position** and **1.8° orientation accuracy** at 450 ms latency by building a **ROS2** 3D perception pipeline using **Open3D** and **ICP** on Kinect v2 point clouds, integrated with an **xArm** for 6-DOF manipulation.

3-DOF Teleoperated Robotic Arm

Feb – May 2025

- Reduced control latency to **85 ms** end-to-end with **<3°** angular error by mapping **MediaPipe** full-body pose estimation to servo joint commands published over **ROS2**.

Vision-Based Autonomous Patrol UAV — Tello Robomaster TT

Feb – Jun 2025

- Built a fully autonomous patrol drone with real-time person detection by combining **OpenCV** vision, a multi-threaded flight architecture, and a **state machine** controller via **djitellopy** SDK.

SKILLS

Programming	C++, Python, C, MATLAB, C#
Robotics & Sim	ROS1, ROS2, Gazebo, NVIDIA Isaac Sim, FreeRTOS, SLAM, EKF, YASMIN, djitellopy
Vision & Percep.	OpenCV, YOLO, MediaPipe, Open3D, ICP, ArUco, point cloud processing
Hardware & Proto.	STM32, Raspberry Pi, xArm, LiDAR, Arduino — CAN, SPI, UART, I2C, TCP/IP
DevOps & Tools	Git, Docker, Flask, VS Code, Linux
Languages	Spanish (Native) · English B2 (TOEFL IBT Jan 2025)

CERTIFICATIONS

TOEFL IBT — English B2 Certificate

Jan 2025